Soft-Bin Lookup for the Discrete-Price Partition Equilibrium at $(\tau, \gamma) = (2.0, 0.01)$, $G = 21$

REZN project, $K = 3$, CRRA, no noise traders

1 What changes

In the base solver the partition-Bayes lookup

$$A_v^{(k)}(m, \mathrm{idx}) \;=\; \sum_{\substack{\text{cells} \\ \text{assigned to } m}} f_v(\text{other signals})$$

is built with a *hard mask*: cell (i, j, l) contributes to a single m (its assigned price level). This makes $A_v^{(k)}(m, \cdot)$ a noisy, jagged function of m for fixed idx, and gives stretches of oscillation in K-means iteration.

Soft-bin replacement. Replace the hard mask by a Gaussian kernel weight indexed by the *discrete* price levels:

$$w_{(i,j,l)}(m) \;=\; \frac{K_{h_p}\big(P(u_{ijl}) - p_m\big)}{\sum_{m'} K_{h_p}\big(P(u_{ijl}) - p_{m'}\big)}, \qquad K_{h_p}(x) = e^{-x^2/(2h_p^2)}.$$

Each cell now contributes a smooth weight to multiple levels, so the lookup $A_v^{(k)}(m, \mathrm{idx})$ varies smoothly in m; the posterior $\mu_k(u)$ becomes smooth in u even though the equilibrium price $P(u)$ is still projected onto the discrete grid each iteration. We use $h_p = 1.5\,\Delta p$ (a constant $\sim$one and a half grid spacings).

2 Numerical comparison

Table 1: HARD vs SOFT at $G = 21$, $\tau = 2.0$. Period = 1 is converged; ≥ 2 is a stable cycle (tail-mean reported).

	period		deficit (tail mean)		max residual	
M	HARD	SOFT	HARD	SOFT	HARD	SOFT
16	1	1	0.198	0.205	4.99e-2	4.99e-2
32	1	1	0.178	0.186	5.00e-2	4.99e-2
64	**2**	**4**	0.196	**0.174**	4.99e-2	4.99e-2

What the numbers say. Max residual is identical in both — it floors at $\frac{1}{2}\Delta p$ either way, because P is still projected to the grid. Deficit values agree within a few thousandths at $M \in \{16, 32\}$. At $M = 64$ the soft lookup gives a meaningfully lower deficit (0.174 vs 0.196), even though both still cycle.

3 What it does to the partition geometry

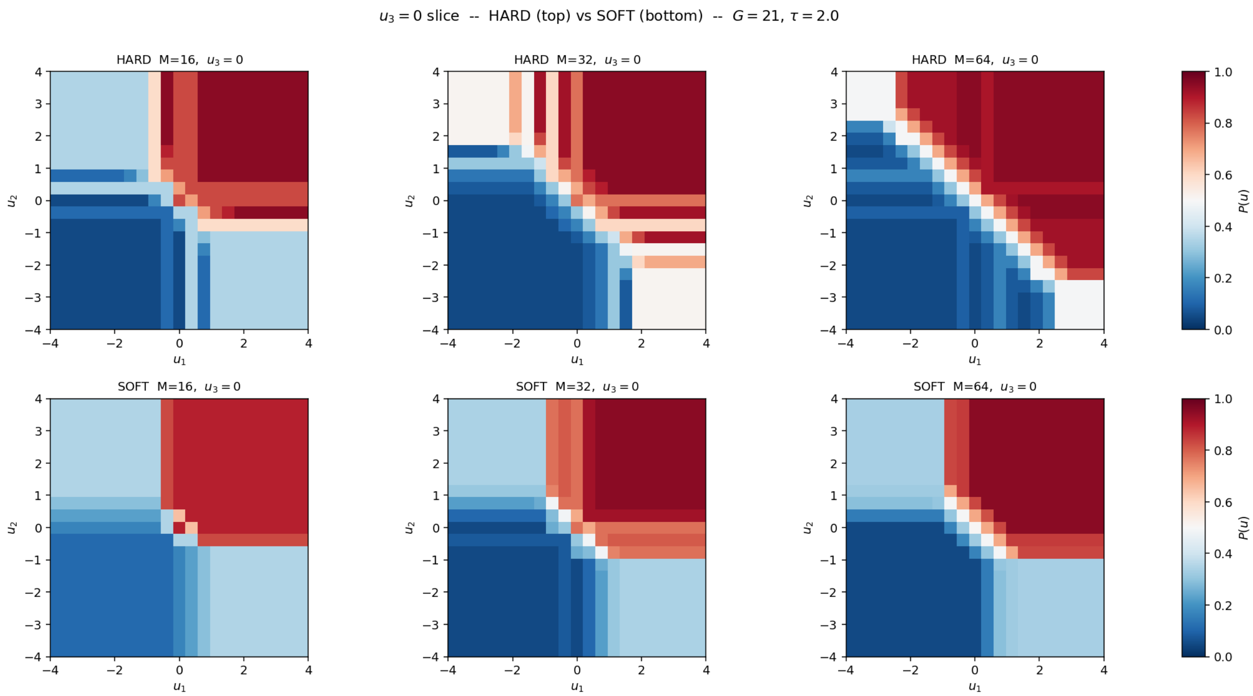

Figure 1: $u_3 = 0$ slice of $P(u_1, u_2 \mid u_3{=}0)$ at $G{=}21$. Top row: HARD lookup. Bottom row: SOFT lookup ($h_p = 1.5\Delta p$). The soft lookup produces a visibly cleaner partition: HARD at $M = 64$ shows scattered speckle along the diagonal (the periodic-2 instability plays out as cells flip between two adjacent levels), while SOFT at $M = 64$ has a smoothly graded transition zone and monotone level sets. This is the regularising effect of the kernel weight: cells near a level boundary contribute to both neighbours, so the partition geometry stops fragmenting.

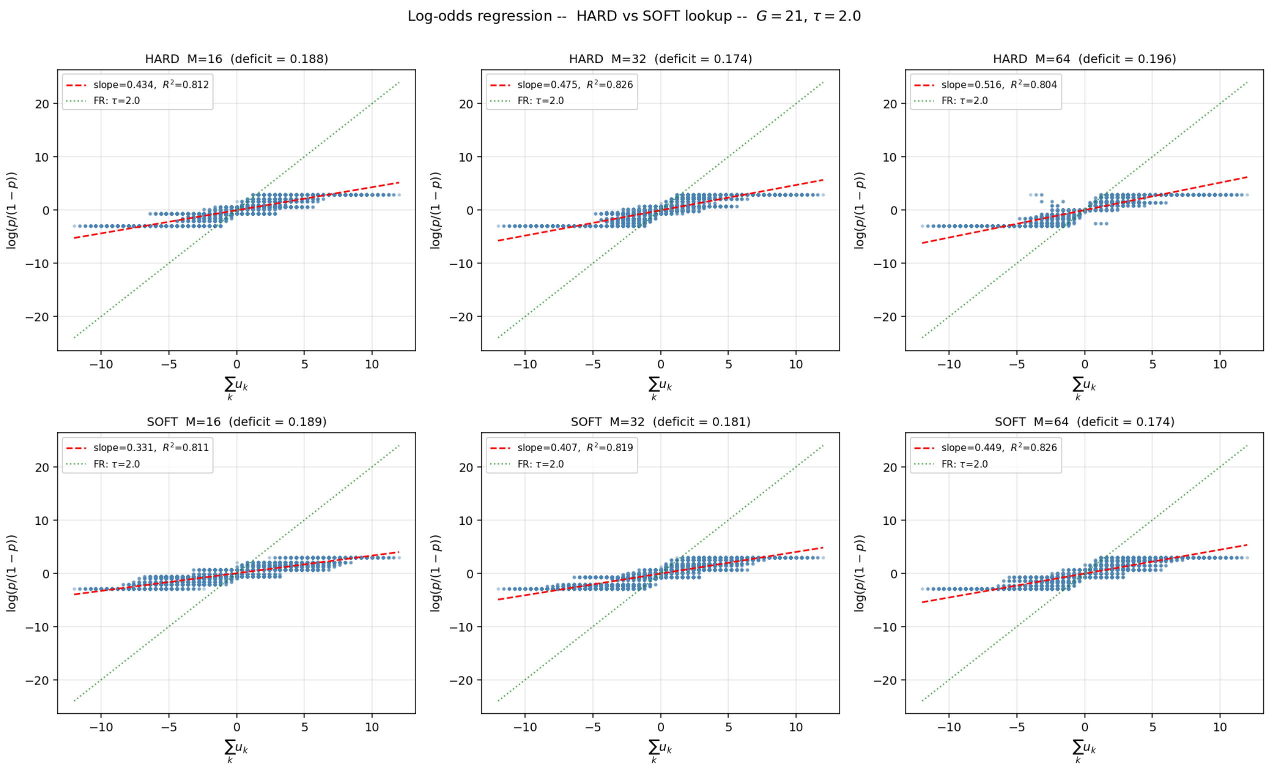

Figure 2: Log-odds regression at G=21, τ=2.0. Top: HARD; bottom: SOFT. Both equilibria are still discrete-valued, so the points band by p_m in both rows — but the SOFT regression line is fit to a less-fragmented field, and at M=64 has R^2=0.826 vs HARD R^2=0.804.

4 Takeaways

- **Lookup smoothness is decoupled from field discreteness.** $A_v^{(k)}(m, \mathrm{idx})$ becomes a smooth function of m while $P(u)$ remains on the discrete grid $\mathcal{P}$.
- **Geometry is the win.** Soft binning reorganises the partition into clean, monotone level sets. HARD at M=64 had a fragmented diagonal; SOFT cleans it up.
- **Deficit is preserved or slightly improved.** No regime where soft loses meaningfully; at M=64 soft is ~ 0.02 lower in deficit.
- **Cycling is not fully cured.** Going from period-2 to period-4 at M=64 shows the cycle is still present, but the orbit is now over a cleaner family of fields. Annealing $h_p \to 0$ might close it; left for later.

Data & code.

- Solver: `projects/REZN/solver_code/cmm_endogenous/discrete_price_soft.py`
- Run: `TAU=2.0 OUT=...lowtau/discrete_price_soft_tau2 python3 discrete_price_soft.py`
- Results: `projects/REZN/solved_fixed_points/lowtau/discrete_price_soft_tau2/`